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Artificial Intelligence and the Application of Deep Learning in the Production of Cadastral Maps

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1. Introduction

Abstract

With the advancement of computers in the present era and the interaction and use of computers, the use of an increasing volume of data, the need for data mining and intelligent software and hardware tools is felt more than ever. Despite the automation of structures and the acceleration of processes, human decision-making and wisdom in guiding these systems as observers and guides of the complex are inevitable. Therefore, after the invention of the computer, scientists have always tried. They have developed intelligent systems that can operate independently of human intelligence by using inference in decision-making, data management, and processing. Considering that mapping has also undergone a major transformation after the emergence of spatial information systems with a large volume of this and that issue, which is only digital data has been encountered. Percentage of the world's lands have official cadastral maps and land registration systems [1, 2]. The use of artificial intelligence in the production or analysis of maps is inevitable and at the same time efficient, and its implementation can increase the speed and accuracy of processes and prevent possible human error.

In registration surveying, map preparation is based on sensitivity. Legal and financial aspects of cadastral maps are very costly. And it takes time. On the other hand, digitizing maps Old paper and production or updating of cadastral maps Aerial photography is a challenging task in this field that requires the precision of expert professionals. Today With the advancement of computer science in the field of image processing, technologies in the field of artificial intelligence have provided us with the possibility of machine learning using algorithms such as neural networks. In this article, we will review Mixed the definitions of these two fields and the measures taken throughout the world. We will examine the progress made in the world using artificial intelligence in the production of cadastral maps. Finally, we will compare the performance of machines and humans in this field. Background, suggestions for better use of intelligence technologies We will present artificial in cadastral mapping. **Keywords:** Cadastre, land administration systems, artificial intelligence, deep learning, convolutional neural network.

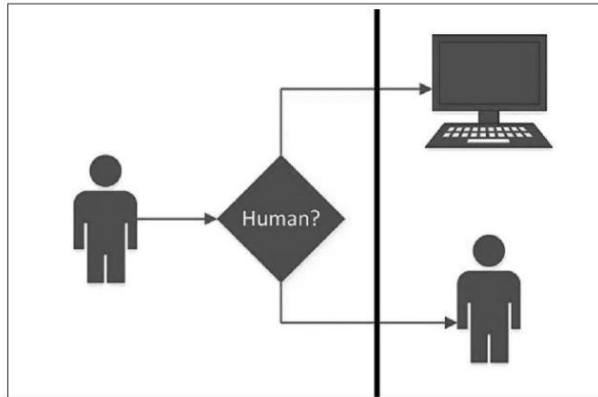


Figure 1: Turing test [13]

-1-2-2 Turing Test Before

the invention of computers, humans always tried to simulate natural intelligence by building mechanical robots.

But the first work in the field of artificial intelligence was done by Alan Turing.

He was from England in the 20th century. He first proposed the idea of testing machine intelligence [10], in which

Problems that require human intelligence to be solved are posed to two anonymous respondents by a person acting as an experiment observer [13]. (Figure 1)

The respondents, one human and the other computer equipped with artificial intelligence, examine the question and answer it. The questioner, who considers the identity of the computer and the human to be hidden, examines the answer. If the questioner cannot determine from the results of the answers which respondent was human and which was machine, it can be assumed that the analytical platform used in the said machine is artificially intelligent.

It was artificial [10].

-2-2-2 Types of artificial intelligence in terms of power

Despite the different methods in artificial intelligence

Scientists generally divide this technology into the following two categories:

They divide.

³ -1-2-2-2 Strong or general artificial intelligence

In this system, the device is equipped with several intelligence models.

Turing Alan 2-
AI Strong 3-

2-Definitions

-1-2 Register

The cadastre or boundary map can be viewed as an ordered list.

A collection of information about belongings within a given area or

A specific country is defined based on the mapping of its borders and

boundaries. Such properties are systematically identified with unique identifiers [3, 4]. This system includes all land information, including

geometric, structural, user, technical and legal characteristics, which

is expressed by recording information on maps, among its applications
can be used

In urban management, resolving legal and judicial issues, economic applications, its application in the environment and natural resource

management [5]. However, the most

important advantage of using the cadastre is the reduction of fair

taxation from the people [6]. Legal claims and in fact the cadastre is

the basis and foundation of land management and development [7,

8]. A proper cadastral system supports the provision of property

rights and the mobilization of land capital, without which many of the

development goals of countries cannot be achieved or

Or they face severe problems [9].

-2-2 Artificial Intelligence

It refers to a software capability in which the ability to perform processes that only humans are capable of performing is designed and implemented. So that

The machine can operate independently in the relevant field.

Humans can make decisions and take action [10,11,12]. This ability

is usually used in system development projects that use intelligent analytics to deploy and simulate capabilities.

It is specific to humans and is implemented in computers. Including

These abilities include inference, finding meaning, generalizing

content, and learning from experiences [10].



Figure 2: Types of machine learning methods [15]

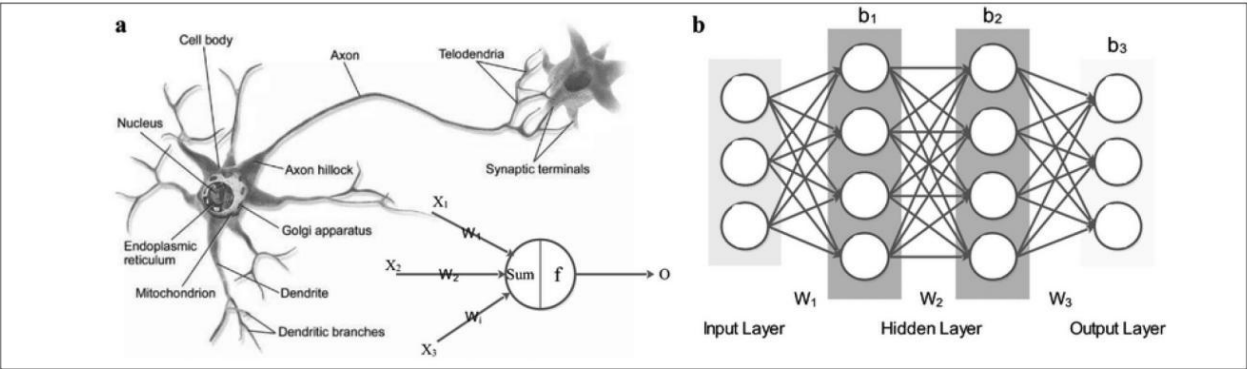


Figure 3: Natural and artificial neural network configuration [16]

It is possible to do this without the need to code complex analytical algorithms [14]. These learning processes are performed in supervised, unsupervised, and semi-supervised forms, as follows [15]. (Figure 2)

1.3.2.2 Supervised learning

In this method, data is classified and labeled by the system administrator to the model so that it can classify other data through cognitive analysis [14].

-2-3-2-2 Unsupervised learning

In unsupervised learning, all input data to the model is also considered as training data, and the machine independently classifies them by finding possible groups [14].

-3-3-2-2 Semi-supervised learning

This method is a combination of the two previous methods. That is, while the computer receives data as training input, it also

It is artificial and can perform various processes simultaneously. Who is capable of playing Game of Thrones? Like the AlphaGo system. Play 5 [12]. Go

-2-2-2-2 Weak or limited artificial intelligence

In this type of artificial intelligence, the system is programmed to perform only one specific task [12]. This model is It is a commercial application and is also known as advanced information processing. For example, medical diagnosis systems and stock trading systems are among these [10].

-3-2-2 Machine Learning

Machine learning refers to a set of models by which a computer can be trained to perform analytical and intelligent processes that only humans can perform, so that it can perform the task as an independent device , perhaps with greater accuracy, than humans. The goal of machine learning is to equip computers with intelligence.

Computer	AlphaGo 4-
Supervised 8-	game board Go 5-
Unsupervised 9-	AI Applied 6-
supervised-Semi 10-	Training 7-

Topographic mapping does not meet the required expectations, because it may lead to inferior results compared to human vision due to the lack of an explicit topology for the objects [26]. However, unlike the first method, the objects resulting from the analysis Object-based images have topological features, such as shape and size [27], which makes them suitable for extracting cadastral boundaries [21, 24, 28, 29].

3- Applying artificial intelligence in cadastre

-1-3 Automatic conversion of cadastral map images
Paper to digital maps

One of the applications of using machine vision algorithms in mapping is converting analog maps into Digital vector data. Through the steps taken, it is possible to draw cadastral maps from scanned images of maps, including identifying and extracting features such as properties, their property handwritten texts on identification codes, and recognizing the map [30]. (Figure 5)

In this process, first the noise of the scanned images is obtained using the non-local averaging method [31], then the SLIC method is used to extract the drawn property lines. [30] which uses clusters of pixels, It generates and extracts superpixels¹⁵ that have the same similarities, are adjacent to each other, and have reduced complexity [32, 30]. These superpixels are merged together and form regions that are divided into three categories of text, rate curve, and background using the support vector machine method¹⁶ [30]. Next, a filter designed for extracting vessels from images¹⁷ [33] is used to extract the parcel and

At the same time, it also processes handwritten texts in images.

means local-Non 14-
pixel Super 15-
(SVM (machine vector-Support 16- filter
enhancement Vessel 17-

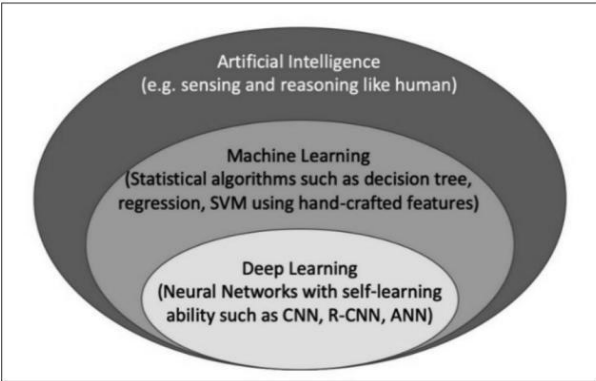


Figure 4: The relationship between artificial intelligence, machine learning and deep learning [17]

Classification receives the rest of the data for learning. It self-categorizes [15].

-4-2-2 Deep learning with artificial neural networks

One of the machine learning methods is called deep learning, which simulates the brain's neural network to learn from and perform the desired task [16]. (Figures 3 and 4) Data

-5-2 Smart Register

The concept of smart cadastre goes back to the field of spatial intelligence in Howard Gardner's theory of multiple intelligences [18,19,20,21]. In which spatial intelligence is defined as the ability to has defined the visual-spatial perception of the world [20, 21]. As this understanding includes the ability to visualize complications Geographical and also the location of these features [21]. In the field of cadastre, the ability to acquire and use spatial intelligence in detecting and extracting property boundaries is referred to as cadastral intelligence [22, 23].

This automatic extraction of cadastral boundaries can be divided into two categories: (a) pixel-based approaches¹² and (b) object-based approaches, which are referred to as Object-based image analysis is also known as [13], [21] [24]. The first approach considers only the spectral value or one aspect for the boundary layer [21, 25]. This approach is used in applications

Gardner Howard 11-
(PBA (Approaches Based-Pixel 12-
(OBIA (Analysis Image Based-Object 13-

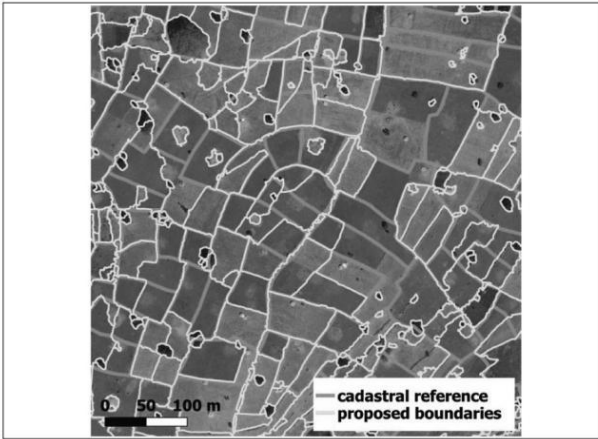


Figure 6: Cadastral map extracted by the mentioned method and map [34]
Reference register

3D buildings in big cities have images

The dense concentration of buildings, the presence of shadows and plants throughout the city make the production of maps in this area very costly and It has taken time [36].

There are many methods for creating 3D models of buildings, including:

From Elidar point cloud data, remote sensing data or mentioned aerial photographs [36, 37].

However, in this study, which aimed to reconstruct

Buildings in three dimensions for use in

Land management in the Tehran metropolis, through the presented steps (Table 1), the results have shown that a three-dimensional model can be

It can measure the building with an accuracy of better than 60 centimeters.

Initially, the aerial analog images of Tehran city were

acquired using the Leica DSW 700 aerial photo film scanner.

Digitized, with the preparation of internal justification parameters

That is, assigning fiducial position to brands, focal length

Camera, datum height and lens distortion of the imaging camera,

as well as preparation in the INPHO software environment of

external reference parameters, which include the use of ground control points (dual-frequency GNSS satellite positioning), the

points and drift processing based on use of survey

triangulation prepared in the original images, pre-processing and

GCP - Points Check Ground 19-

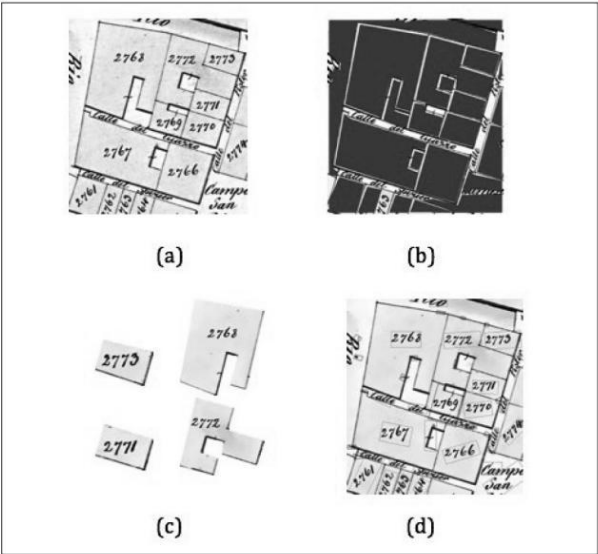


Figure 5: Sample result a) Main map b) Polygonal property estimation

c) Extracted properties d) Identifying the range of property identification codes [30]

It works with convolutional neural networks to extract the

property identification code [30].

-2-3 Semi-automatic conversion of images into maps

Register

In this method, both the artificial intelligence agent and the user are involved in converting drone images into cadastral vector maps. [34], which was designed in three stages.

(a) Digitization of closed curved lines from features

Visible in drone images using a hybrid clustering and multi-

definition method.18 (b) Classification of the extracted boundaries

using a machine learning algorithm so that after the learning process, the computer can identify which boundaries are efficient

for the cadastral map and finally (c) where the user creates the

polygon shape of each property by selecting adjacent

It is.

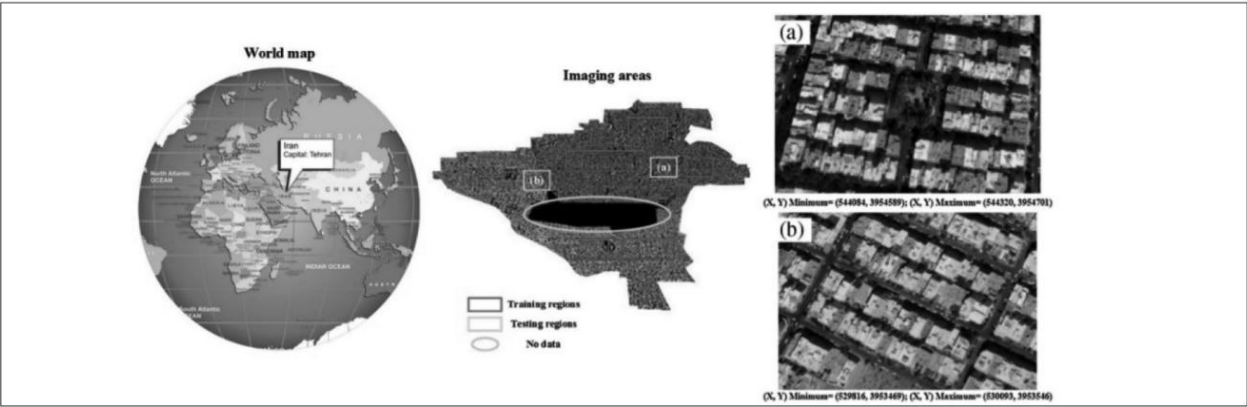
boundaries using a QGIS software plugin [34, 35]. (Figure 6)

-3-3 Modeling 3D cadastral maps

By deep learning

One of the biggest problems in building models

() grouping combinatorial Multiresolution 18-



(b) Part of Sadeghieh neighborhoodFigure 7: Study area (a) Part of Resalat neighborhood and

It checks the accuracy of the data [36], in which in this process the matrix categories of pixel errors are classified and checked for two categories: building and non-building. [36].

Study scope ²¹ Next, using the image pair to calculate the height of the buildings by the error caused by the height difference, the height of the buildings is calculated and using the obtained parcels from the deep learning model (results of the previous stage), a 3D model of the buildings is prepared [36].

(Figure 9)

-4-3 Identifying possible cases of land grabbing with Using artificial intelligence and cadastral data

Since studies have shown that in many places in the Netherlands, there are many cases of

There is land grabbing [38], in this method the model designed from the drone images is 22 vertical compared to the feature separation 23

Using a deep neural network model and generating maps of land use classifications²⁴ , they were then compared with existing cadastral maps and land use maps.

Public and private, parts that are likely to be encroached upon and identifies the owner's progress on public lands. (Figure 10)

stereo-Geo 21-
photos Ortho 22-
Features 23-
Classification Use Land 24-

Table 1: Steps of 3D cadastral modeling using aerial images by deep learning-based methods [36]

First stage
Collecting aerial images
Preparing internal justification parameters
Preparing external justification parameters
Second stage
Automated interpretation and extraction of buildings using aerial images and methods
1FAIDL Deep learning
Third stage
2GSM and 3D modeling of buildings using FAIDL

learning deep with interpretation automatic Fully 1- mapping stereo-Geo 2- [36] For this study, two parts of Tehran city were selected. Selected from the neighborhoods of Raslat and Sadeghieh, and parts of the city were also selected as training data for the deep learning model [36]. (Figure 7)

Digital images prepared using fully automatic deep learning interpretation of the SegNet model (Figure 8), which has the ability to extract edges and object identity despite the complexity of the images, have been applied in the MATLAB a2017 software platform using the MatConvNet tool.

[36]. For better coding, we also use hand normalization.

The ReLU operator was used and finally the model was processed using a Geforce Ti 1080 GTX graphics card [36].

To evaluate the results, the accuracy check process is carried out with the map. 1:1000 is designed by creating a matrix of numbers, normalization Batch 20-

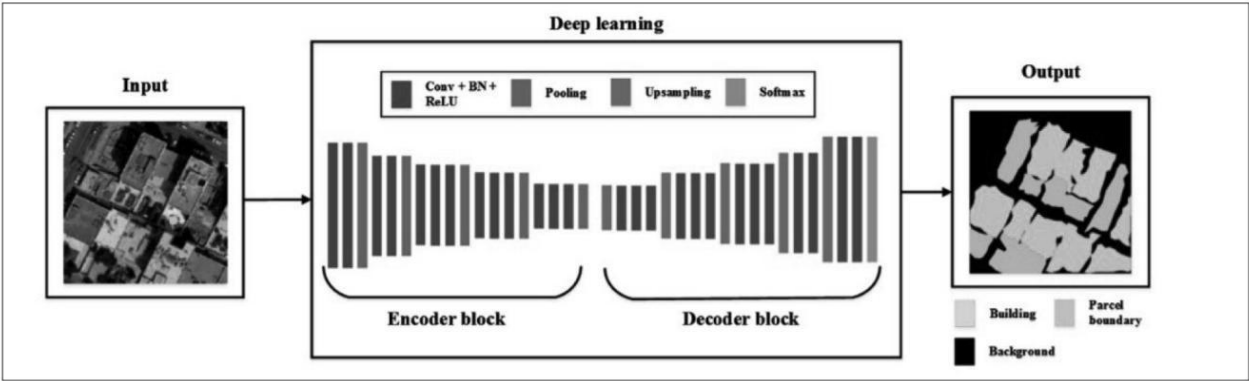


Figure 8: Deep learning structure and example of input and output [36]

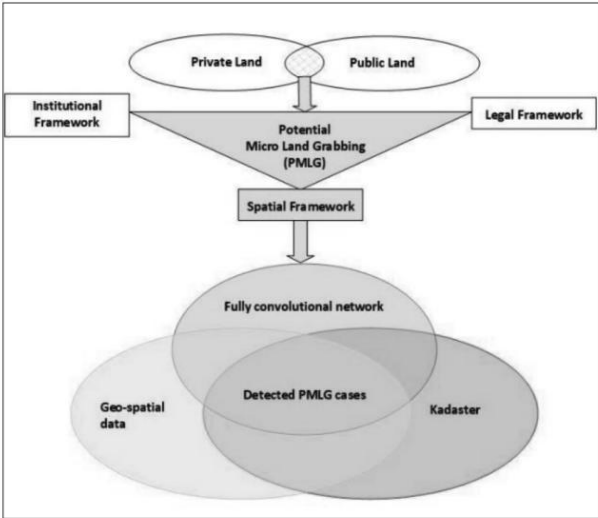


Figure 10: Conceptual map of small areas with potential for land grabbing Aerial imagery using deep learning

(PMLG (Grabbing Land Micro Potential 1- Two different types of data not only increase the learning potential of the model, but also enrich and diversify the classifiers in the image [39].

In the city of Zawala, 26 aerial image mosaics are Randomly selected mosaics, each with a size of 2500 x 2500 pixels and a spatial resolution of 0.1 m, 18 mosaics of these images were used as model learning data and 8 mosaics were used to test the results.

[39] (Figure 11)

In the city of Zhytomyr, color images in dimensions of 2560 Pixels with a spatial resolution of 0.1 m were used, resulting in a total of 18 mosaics, one in each, including 9 mosaics for mosaics for learning the selection model. testing and 9

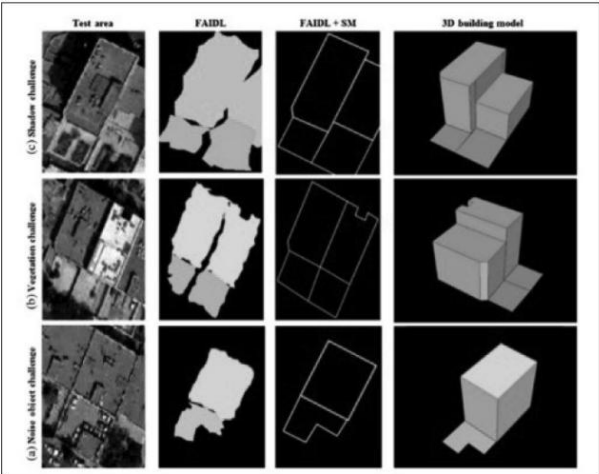


Figure 9: Reconstruction of the scene and the model resulting from deep learning and its 3D modeling using height [36]

In this study, a fully connected fusion deep learning model with an extended kernel was used [39].

This model uses kernels to connect all hidden layers and resize pixels by spacing them [39]. This study consists of 7 hidden fusion layers, with and Enlarge the image. integrated normalization and the LeakyReLU activation function, a fusion layer for classification, a deletion layer for over-occurrence reduction, and a maximum smoothing layer for building 26

The output is classification [39].

For example, the two cities of Zavala27 and Zhytomyr28 in the country The Netherlands was chosen as the study area, despite

(DKs-FCN (Kernels Dilated with Network Connected Fully 25- SoftMax 26- Zwolle 27- Zoetermeer 28-

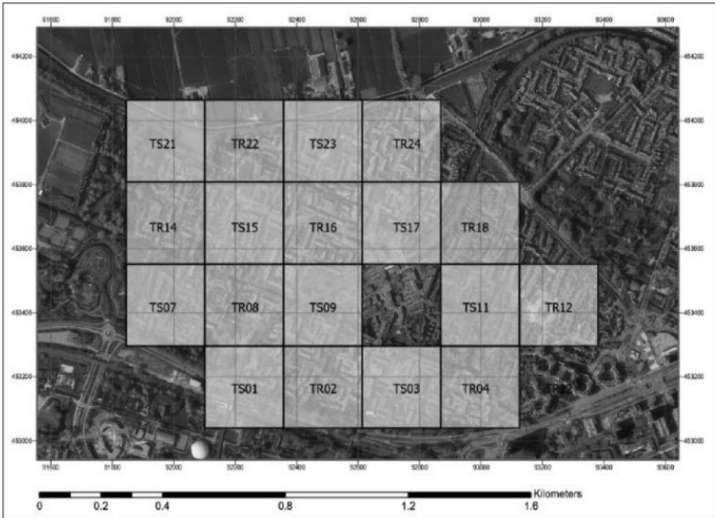


Figure 12: Learning and assessment areas from the city of Zhytomyr selected one by one. Blue: learning, yellow: assessment [39]

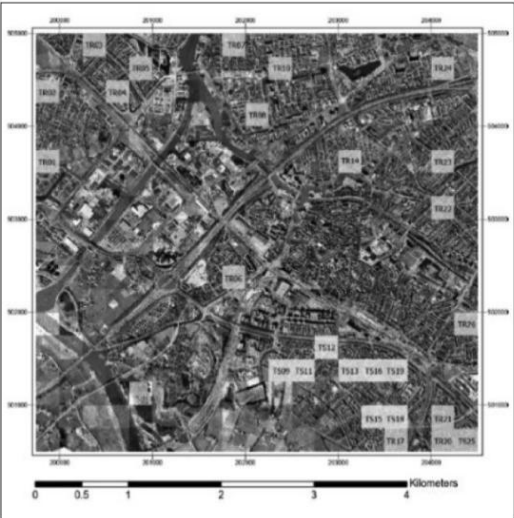


Figure 11: Learning and assessment areas from the city of Zawala, randomly selected. Blue: learning, yellow: assessment [39]

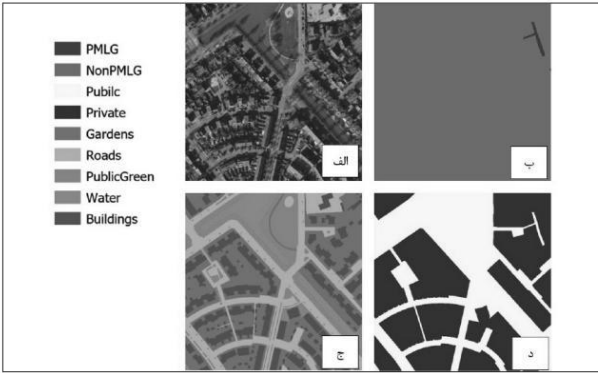


Figure 13: Spatial information of the city of Zawala Mosaic 15-A (Color photo)
b) Manual digital data c) Land use and cover reference data
D) Land rights data (public and private)

Once 56% and again 58% of the two layers matched, revealing identical pixels from the PMLG.

They showed [39].

4-Conclusion

1.4 Results of evaluation and comparison of models

Reviewed in these study evaluation methods
The paper has calculated the following formulas and is presented in Table 2 as a comparison of these models. These values indicate the coverage (Formula 1), accuracy (Formula 2), precision (Formula 3), and the share of two sets on their community (Formula 4).

This process is carried out in two stages. First, using the SegNet deep learning fusion model land cover²⁹ and land use³⁰ maps are generated, then compared with Land registry maps³¹ are divided into two categories: personal and Generalization and application of the 6DK-FCN deep learning model
Binary land-grabbing maps are extracted, and the accuracy of this method is evaluated on three experimental mosaics.
Selected 86.7%, 82.0% and 83.6% calculated

Process of identifying potential land grabbing using and Deep learning models for two-stage LULC classification
PMLG diagnosis is divided [39]. By comparing images, many cases of land grabbing can be identified.
which are present in both the cadastral layer and the layer resulting from artificial intelligence analysis. Since the cadastral layers are also They are not 100% accurate, you can't be sure.
The missing items were not in the manual digitization of PMLG.
And this processed data also requires field visits.
[39]. But by examining the pixels of both outputs, in general

map Cover Land 29-
map Use Land 30-
(land Public/Private (map Rights Land 31-

Table 2: Comparison of model evaluation results

۳۷ صحت	۳۶ پوشش	دقت	۳۵ IoU	تبدیل خودکار تصاویر نقشه‌های کاداستر کاغذی به نقشه‌های رقومی [30]
0.75	0.72		0.6 <	
0.71	0.69		0.7 <	
0.62	0.60		0.8 <	تبدیل نیمه خودکار تصاویر به نقشه‌های کاداستر [34]
0.76		0.52		
0.90	0.94		0.83	مدل‌سازی نقشه‌های کاداستر سه‌بعدی به‌وسیله
0.92	0.95		0.82	یادگیری عمیق [36]
0.16	0.56		0.56	شناسایی موارد محتمل بر زمین‌خواری با استفاده از
0.23	0.42		0.58	هوش مصنوعی و داده‌های کاداستر [39]
				LULC_ex3_RA
				LULC_ex5_RA

The intersection of two sets over their intersection – Union over Intersection 35.
Completeness or Recall 36.
Precision 37.

According to the studies conducted, it can be concluded that

that these models have significant potential for progress towards

Fully automatic generation of cadastral maps. In cases where

area we are dealing with large volumes of data and a large

(such as a country), high processing speed in image

digitization is very important.

which justified the lack of accuracy in the maps produced

Because a large part of a country's assets are

It is related to the agricultural cadastre and, while taking into account the cost

Mapping with one of the high-resolution methods, this

The maps will not require the precision and sensitivity of an urban

the other hand, as has been seen, the use of artificial intelligence

Artificial for real estate plots that do not have buildings and easements

are urban, a much more favorable outcome for the cadastre

It will include an agricultural versus an urban cadastre.

-3-4 Suggestions

Studies conducted on the use and According to the measures

Using artificial intelligence for image processing in mapping

and cadastre are important suggestions that we will discuss in this

section. First, it is suggested that for the application of deep learning

algorithms, due to the high volume of data processing, processing on

the computer's graphics card should be used instead of its main

processor. Regarding the architecture of the artificial neural network

Computer

algorithm, as seen, the use of the SegNet deep learning model should

1401 be considered in view of the structure

$$\text{مثبت درست} = \frac{\text{مثبت درست}}{\text{مثبت درست} + \text{منفی غلط}}$$
 Formula 1 [36]:

$$\text{صحت} = \frac{\text{مثبت درست}}{\text{مثبت درست} + \text{مثبت غلط}}$$
 Formula 2 [36]:

$$\text{دقت} = \frac{\text{مثبت درست}}{\text{داده های مرجع} + \text{داده های پیش بینی}}$$
 Formula 3 [39]:

$$\text{مثبت درست} = \frac{\text{مثبت درست}}{\text{اشراک دو مجموعه بر اجتماع همه نتایج}}$$
 Formula 4 [39]:

-2.4 Differences in the performance of experts and learning models

Car

From comparing the digital borders manually by

Experts identified deviations from automatically generated rural land cadastre. On

boundaries.

The automatically generated features were very few. In the extracted

property polygons, So that almost

Less than 20% of areas have undergone changes or removal of a

[21]. In fact, it can be said that human operators are more accurate than

machine algorithms when drawing and reproducing the geometry of

shapes. However, machine performance is very high in rural data

contexts due to speed and volume [21]. However, in urban areas,

humans are better than automated systems.

And the buildings performed better and the property boundaries

The machine is produced in terms of topology and structure geometry.

And buildings and comparable properties had poor

They were not digitized by hand [21].

://https :Available .]Online [2021. November 09 ,Encyclopedia Ac
[.intelligence-artificial/technology/com.britannica.www .]2021
November 14 cessed-Over

“,Vukoman ,Jokanoviĭ ,Garg Kumar Pradeep] 11[
,Intelligence Artificial in” ,Intelligence Artificial of view-4. .p 2022,
10.1201/9781003140351-1,
your s'lt - Tutorial Intelligence Artificial “,McCarthy .J] 12[
November 27 ,Team DataFlair” ,future the innovate to time
arti/blogs/training.flair-data://https :Available .]Online
[2019. /.tutorial-ai-intelligence-
ficial-Moham Murtala ,Maitalata Sanusi Umar ,Haruna Usman] 13[
Chatbot Intelligence Hausa “ , Maitama Zubairu
Jaafar ,med-206-219. .pp 2021, 978-3-030-69142-4,=isbn” ,System
Rüdiger .Dr ,Reichert Manfred .Dr .Prof ,Zarei Shima] 14[
University of Ulm, Learning Machine, Klaus Kammerer, Pryss

2017. ,sity-,Hasan .T Ramadan ,Hasan Abas Dathar ,Zeebaree Diyar] 15[
Machine”, Ahmed HY Falah, Abdulazeez Mohsin Adnan

A :Selection Gene for Algorithms Supervised-Semi Learning
System on Conference International th11 IEEE in” ,Review
2021. ,Malaysia ,JICSET (Technology and Engineering
recommendation intelligent An “,Gao Shu ,Jiang Zilong] 16[
neural deep hybrid on based advertising online for approach
Comput Cluster - Springer” ,computing parallel and network
2020. ,ing-
Fran ,Attia .Z Itzhak ,Aly .H Yousof ,Akkus Zeynetin] 17[
.A Patricia ,OlsonyArruda .M Adelaide ,JimenezLopez ciscoy
Friedman .A Paul ,Kane .C Garvan ,Pislaru .V Sorin ,Pellikka
Echo Empowered)AI (Intelligence Artificial “,Oh .K Jae and
2021. ,Medicine Clinical of Journal” ,cardiography-
Emergence “,Petersen .C Anne and “,C Marcia ,Linn]
18[A :Ability Spatial in Differences Sex of Characterization
and .doi://https .no 56, .vol ,Development Child” ,Analysis-Meta
1985. 1479-98, .pp /10.2307/1130467,org Be
Child of Encyclopedia .,A .J ,Naglieri; .S ,Goldstein]
19[2011. ,Springer :MA ,Boston ,Development and havior-&
Teaching .,D ,Dickinson; .B ,Campbell .,L ,Campbell]
20[Resources Education ,intelligences multiple through Learning
1992. ,Center Information
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Boundary Cadastral Driven-Machine versus Human paring-2019.
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7/3 Commission FIG in” ,Presented .Cadastral Intelligent The”
Juliens St ,Information Land of Crowdsourcing on Workshop
2015. ,Malta
,Yang .,F ,Nex .,M ,Gerke .,R ,Bennett .,S ,Crommelinck] 24[
extraction feature automatic of Review “,G ,Vosselman; .Y .M
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Based Object and Classification Based Pixel of tiveness-

And the results obtained have produced the best output in this
area so far, and it is proposed to use the ReLU function in it.

Or ReLU Leaky can be used for processing in cells.

Before using the trained model, you must also

maps should be Monitoring and evaluation of the obtained
carried out. For this evaluation, it is necessary to have access to
a part of the digital maps prepared by the cadastre of that area.

The selection of the test area of the results should also be from

mosaics of images that have not been used in the model learning process.

And it is better to choose them randomly.

References : Schwer, S., Ho, J., Cromptvoets, M., Gerke, R., Bennett] 1[
Third Building “,R ,Wayumba;,M ,Chipofya .,A
468 ,ing-,UAVs ,Sketchmaps Smart ,land4Its :Tools Land Generation
Annual in” ,GeoCloud the and ,Extraction Feature Automatic
2017. March ,Poverty and Land on Conference Bank
World .properties and lands unregistered of Valuation “,GLTN]
2[developing for approach enhancement security tenure new a in/
net.gltm.www://http :Available .]Online [2015.” ,countries /2243-
documents-gltm/2-send/jdownloads/component/php.dex-

2015eng brief properties and lands unregistered-of-valuation
Car National Iran, Tehran,” Digitalized “Cadater,” Yousefi.R]
3[26. .p 2004, Center tographic-
Me, Hanzae Akhoundzadeh Mehdi, Khoshboresh Mehdi] 4[
processing oriented-Object “,Sadeghian Saeed ,Hasanlou hdi-Li and
resolution spatial high with images sensing remote of for
building the extracting of aim the with cloudpoint
DAR-2019. ,Journal Science Computing” ,mapping urban a is
Sensing Remote “,Sadeghian Saeid ,Zaeri Fahimeh] 5[Ca
Environmental s'Country the Preparing in Solution New
Environment & Development ,Sustainability” ,Maps dastral-/
20.1001.1.24233846.1400.2.4dor/net.dorl://https .no .)JSDE(2022.
May .1.8, Land
and Cadaster of Conference National “,Sadeghian .S] 6[Property
in” ,Approach Governance-E with Administration 2008. ,Iran of
Organization Registration Deeds and land ,ownership
land of Fundamentals “,Donnelly .J .G] 7[on Committee
Intergovernmental” ,surveying and ,boundaries 2012. ,Mapping &
Surveying In-
Geo :Cadastral ,McLaughlin John ,Yomralioglu Tahsin]
8[2017. ,Springer ,Administration Land in Innovations formation-
sys information land and Cadastral “,Williamson .P .I]
9[:DOI .no ,Surveyor Australian” ,countries developing in tems-1986.
27-43, .pp 10.1080/00050326.1986.10435202,
Online Britannica” ,Intelligence Artificial “,Copeland .B] 10[

meth superpixel art-the-of-state to pared -com superpixels
 2012. 2282,-2274 .p 11, .no 34, .vol ,Transactions of the
 IEEE" ,ods-,Viergever .,L .K ,Vincken .,J .W ,Niessen .,F .A ,Frangij] 33[.
 Medical in" ,filtering enhancement vessel Multiscale .,A .M
 .no ,Intervention Assisted-Computer and Computing Image
 1998. 130-137, pp. , Springer
 Cadas Towards " ,Mila ,Koeva & Sophie ,Crommelinck] 34[
 2019. ,International GIM" ,?Intelligence tral-
 Crommelinck Sophie] 35[,Yang Ying Michael ,Koeva Mila
 Deline for Learning Deep of Application " ,Vosselman George
 Sensing Remote from Boundaries Cadastral Visible of
 ation-2019. ,Sensing Remote" ,Imagery
 Deep, Sadeghian Saeid, Masouleh Khoshboresh Mehdi] 36[
 dimensional-three reconstructing for method based-learning
 Ap of Journal" ,images aerial from models cadastre building
 2019. 19, .p ,Sensing Remote plied-cor
 of model formal A " ,Frank.U Andrew ,Bittner Steffen] 37[
 Urban and Environment ,Computers" ,cadastre a in rectness-2002.
 482,-465 .p ,J5(26 .no ,Systems
 Ex An :Netherlands the in Use Land Illegal " ,Hoops .B] 38[
 Illegal Where and Often How on Study Empirical plorative-Jour Eleconric
 SSRN" ,Netherlands the in Occures Use Land
 2017. .3080849,ssrn/10.2139/org.doi://https .no ,nal-in
 Grabbing Land Micro Potential of Detection " ,Shan .S] 39[
 ,Twente of University" ,Learning Deep using Netherlands the 2020.

Composi Floristic In Image Resolution High Using sification
 "(City Magelang Tidar Gunung :Case Study (Mapping tion-2016. ,Ltd
 Publishing IOP
 image based-Object " ,G ,Hay; .S ,Lang .,T ,Blaschke] 26[
 sens remote driven knowledge for concepts spatial :analysis
 2008. ,Media Business & Science Springer" ,applications ing-
 Perform ,Knowledge :Dystopia Academic " ,Roberts .P] 27[
 .Canterbury of University" ,Education Tertiary and ativity-.doi://
 https .no ,Leadership and Studies Educational of School
 2013. /10.1080/10714413.2013.753757,org
 Seven; .A ,Kwabena .,C ,Lemmen .,R ,Bennett .,D ,Kohli] 28[
 Visible Completely of Comparison Quantitative A " ,J ,bergen-Au towards
 Step A :Images Satellite Using Parcels Cadastral
 world the Surveying 2017. Week Working FIG in" ,tomation-Hel ,reality
 augmented to digitalisation From - tomorrow of
 2017. ,Finland ,zinc-
 Based-Object for Practices Good " ,P ,Bogaert .J ,Radoux] 29[
 2017. 646, .p ,Sensing Remote" ,Assessment Accuracy
 ,Lenardo di Isabella ,Kaplan Frederic ,Oliveira Ares Sofia] 30[
 Premiere in" ,plans cadaster on algorithms Vision Machine"
 Hu Digital of Alliance International the of Conference Annual
 August ,Canada ,Montreal ,)2017 DH (Organizations manities-2017.
 8-11,
 something local-non A " ,M.-J ,Morel and .,B ,Coll .,A ,Buades] 31[
 Confer Society Computer IEEE" ,denoising image for rithm-2005.
 60-65, .pp 2, .vol ,ence-Slic
 " ,P ,Fua .,A ,Lucchi .,K ,Smith .,A ,Shaji .,R ,Achanta] 32[

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